International Institute of Information Technology, Hyderabad

(Deemed to be University)

CE 8.401 Disaster Management (Spring 2026)

<u>End Semester Examination</u>

Maximum marks:100
Time:3 hrs

All the best!

Questions

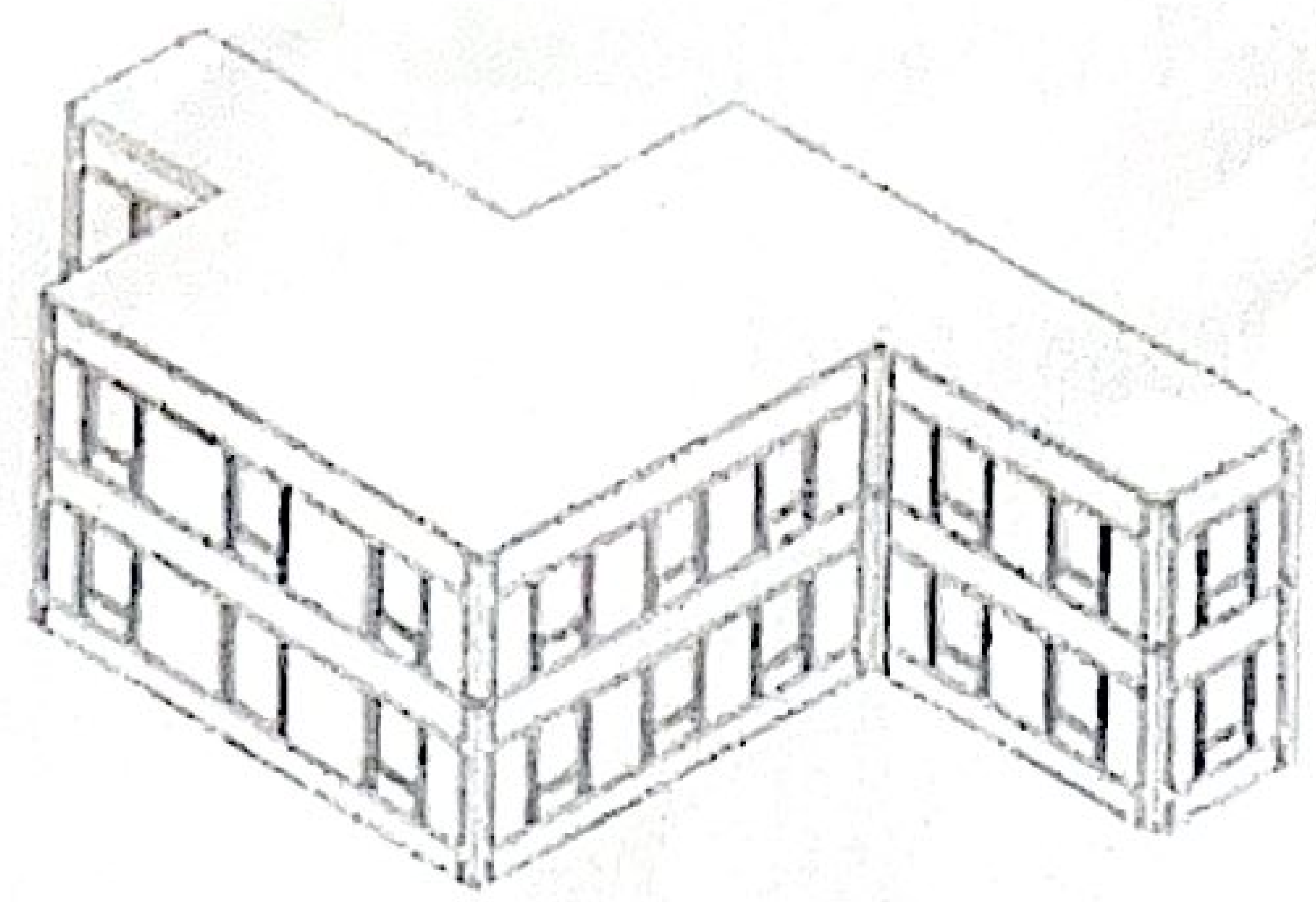
1. You are the lead engineer for a housing project on a 30° slope ground within a region classified as Seismic Zone II. The area is historically arid, but the soil composition consists of loose, unconsolidated debris and weathered sedimentary rock. Which of the following hazards is the most immediate threat to the structural integrity of the building: Flash flood, Earthquake, Landslide, Cyclone? State four measures to mitigate the disaster. ***(5 marks)***

2. A region is identified as high-risk through multi-hazard risk assessment, facing risks from earthquakes, floods, and landslides (high hazard + high exposure + high vulnerability). Seismic data indicates the area lies near an active fault zone, topographic analysis reveals steep slopes prone to mass movements triggered by both rainfall and ground shaking, and hydrological modeling shows recurring inundation from a nearby river system during monsoon seasons. Despite this compounded risk profile, reconstruction of buildings and relocation of the population is not feasible due to economic and social constraints. Suggest mitigation measures for this region to mitigate potential disasters. ***(10 marks)***

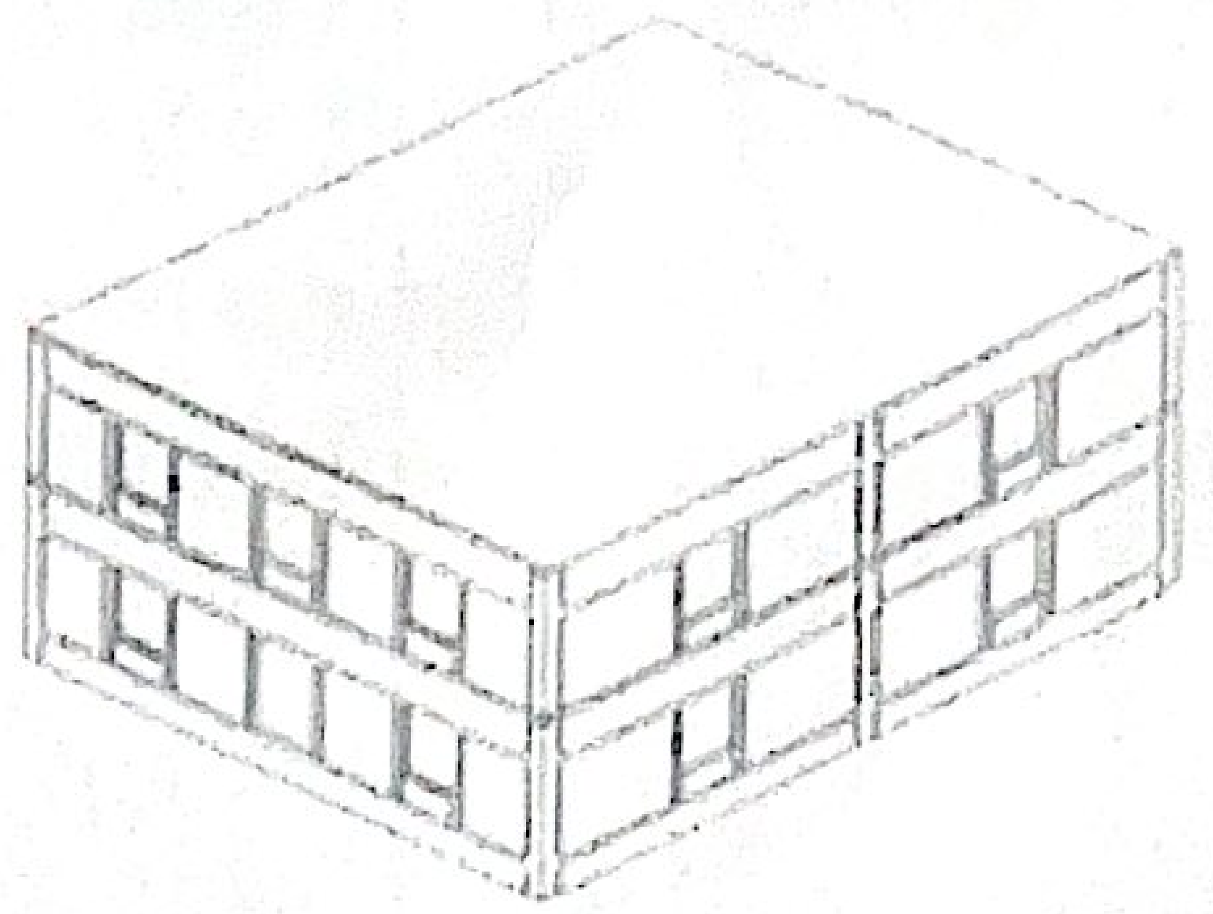
3. 3. An earthquake of magnitude 6.8 strikes a region, causing significant damage to buildings and infrastructure. As a CS//EC engineer working along with a Civil Engineer, you are assigned to participate in the vulnerability assessment of the affected buildings. What measures would you recommend, from a CS/EC engineering perspective to support:
   (i) Response;
   (ii) Recovery and rehabilitation; and
   (iii) mitigation activities in the disaster-affected area? ***(3x5 marks)***

4. (a) How did digital technologies contribute differently to preparedness and response during the 2019–20 Australian bushfires? ***(4 marks)***
   (b) During the 2015 Nepal earthquake, how did digital platforms improve coordination, data collection, and decision-making, and what limitations existed due to the technology level at that time? ***(6 marks)***

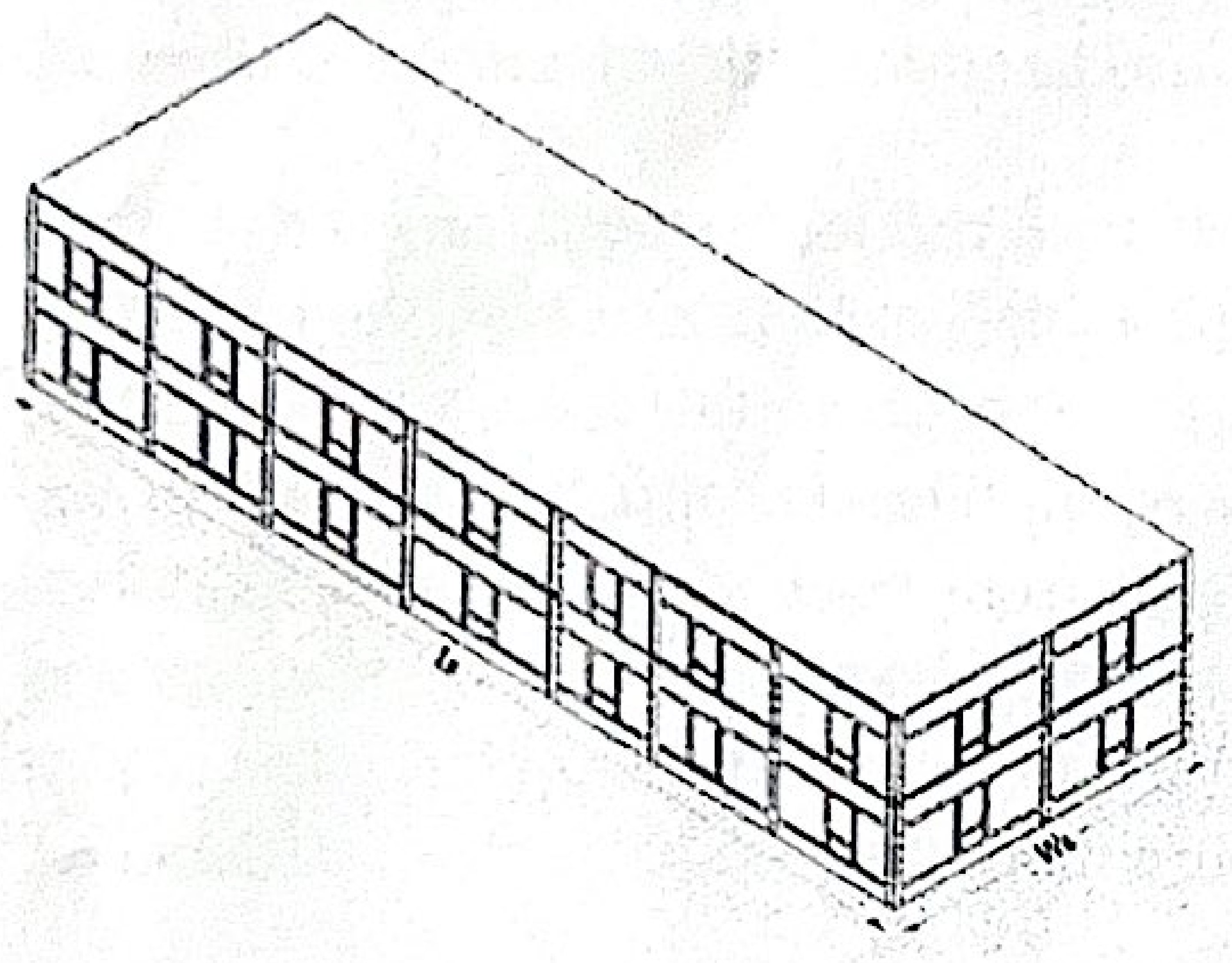
5. You are constructing your house in an earthquake prone region. Which of the following plan configurations will you adopt? Justify. ***(8 marks)***



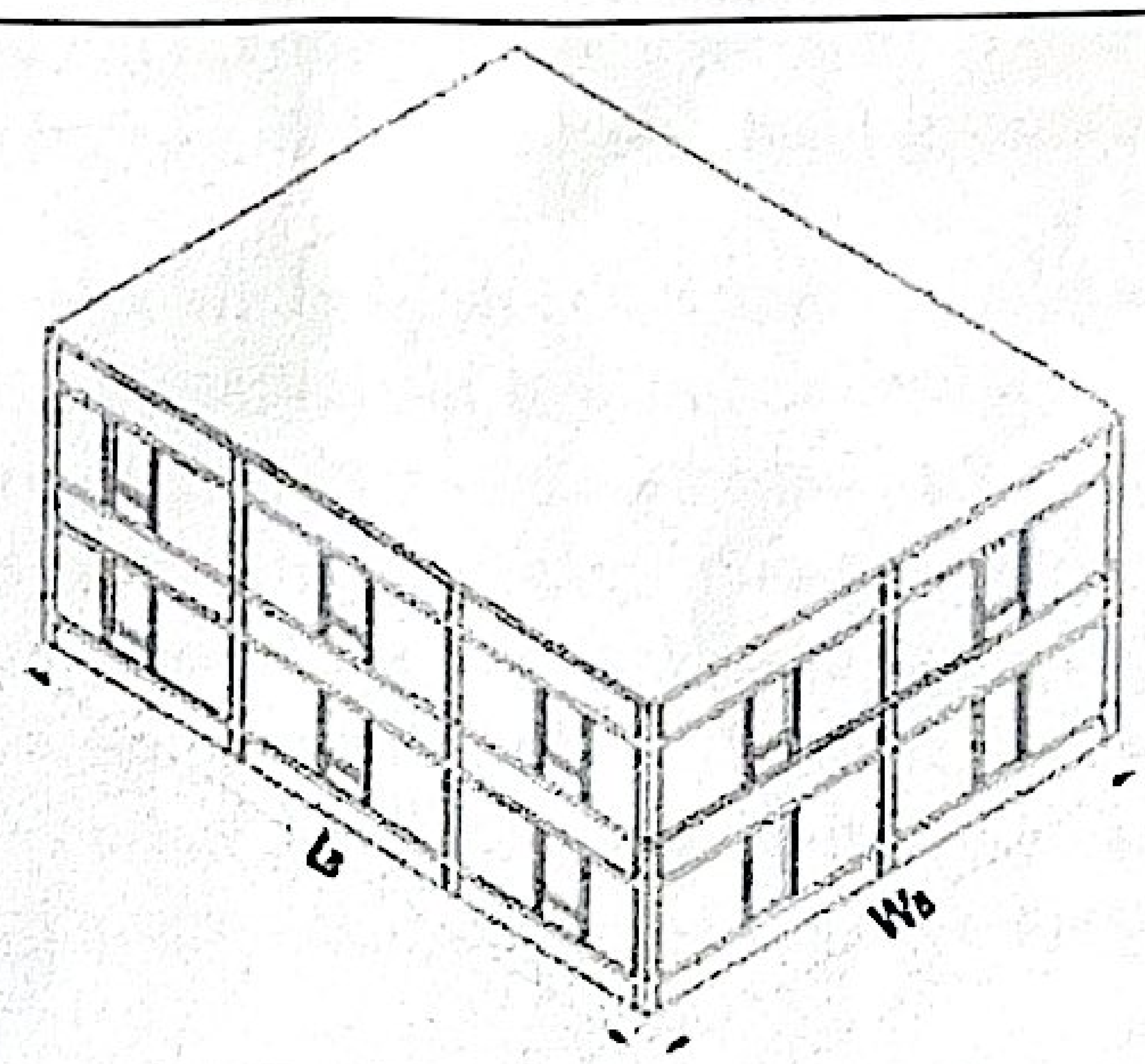
(a)



(b)



(c)



(d)

6. A technology company claims that its new AI-driven satellite monitoring system can "*prevent natural hazards from occurring.*" As a technical reviewer for a funding agency, explain why this claim is scientifically unsound by clearly distinguishing between a Hazard and a Disaster. Provide one concrete example of how such a system could realistically be used to reduce disaster risk, even though it cannot prevent the natural hazard itself. ***(10 marks)***

7. A Category 5 cyclone makes landfall on a coastal city in Peninsular India. As the emergency services are overwhelmed, a cyberattack disables the automated control systems of the regional power grid, cutting electricity to automated flood-pumps and hospital critical care units.
   a) Based on your understanding of complex disaster interactions, classify the above scenario using the appropriate term(s) from the following:
      i) Compound Event
      ii) Cascading Disaster

   Your answer may use one or both terms. Justify your classification with reference to the specific sequence of events described. ***(5 marks)***

b) Why are modern disasters of this type considerably more difficult to manage than a standalone natural hazard? Discuss at least two factors that increase management complexity. *(5 marks)*

8. Tsunami warning sirens are activated along a coastal tourist town. Red Alert messages are broadcast to all mobile phones in the area. However, surveillance footage shows that rather than evacuating to high ground, a significant number of tourists, many of them unfamiliar with tsunami precursor signs, are moving toward the beach, apparently to observe the unusually retreating ocean.

   a) Explain the psychological phenomena of Normalcy Bias and Optimism Bias. In your answer, also briefly distinguish the role of knowledge gaps in the above scenario from cognitive bias and explain why conflating the two can lead to poorly designed public safety campaigns. *(7 marks)*

   b) Explain how either or both of these factors can directly contribute to turning a manageable natural hazard into a large-scale disaster. Suggest one specific communication strategy that authorities could adopt to improve evacuation compliance in such situations. *(5 marks)*

9. A cloud services company operates a data centre near a coastal technology corridor. The Risk Management team is conducting a Quantitative Risk Assessment (QRA) for a specific hazard: a Category 4 storm surge. The following parameters have been established from meteorological data and engineering surveys:

Parameter	Description
Exposure (E)	The total asset value at risk including hardware replacement, data infrastructure, and contractual liability payouts to clients is ₹200 Crores.
Hazard Probability (P)	Historical climate models indicate that a Category 4 storm surge occurs in this region approximately once every 25 years.
Vulnerability (V)	The facility is slightly elevated, but the backup diesel generators are located on the ground floor. If a storm surge occurs, there is a 60% probability (V = 0.60) that floodwater will breach the generator room, resulting in a complete facility shutdown and the full loss of ₹200 Crores.

   a) Calculate the current Annualised Risk (R_0) for the data centre in Crores per year. Use the standard QRA formula: Risk (R) = P × V × E. Show all working clearly, including conversion of the return period to an annual probability. *(5 marks)*

   b) The engineering team proposes installing an automated hydraulic flood barrier system around the generator room, at an annualised cost (installation and maintenance) of ₹1.5 Crores per year. If implemented, this measure would reduce Vulnerability from 0.60 to 0.05. Calculate the new Annualised Risk (R_1). Then, conduct a financial Cost-Benefit Analysis and state whether the company should approve the investment. Support your answer with calculations. *(9 marks)*

   c) The flood barrier system relies on automated water-level sensors to trigger the hydraulic doors. Suppose a cyberattack disables the sensor network at exactly the moment a storm surge is approaching. Using the concept of Cascading or Interconnected Disasters, explain what happens to the Vulnerability (V) and the resulting Risk (R) in this scenario, and what broader lesson this illustrates for infrastructure risk management. *(6 marks)*

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